Two-Degree-of-Freedom Visual Stabilisation System on the PIC18

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Abstract—**This project explores a two-degree-of freedom visual stabilisation system running on the PIC18F87K22. An orientation sensor and pressure sensor give estimates of angle and height. A discrete feedback correction loop corrects the image on the GLCD to achieve visual steadiness for x–y rotation motion and z-axis motion. The objective of the system is to achieve the following three measurable goals: angular drift of less than 20°, a radius error of less than 1-3cm, and a closed-loop update rate of 20-25Hz. The performance of the system will be tested with controlled motion tests and a discrete-time simulation. The design gives a minimal but extendable basis for full 3-DOF motion compensation in future work.**

I. Introduction & Motivation

This project explores a low-dimensional version of a visual stabilization system. The concept is straightforward: when the device moves, the image that is displayed should remain visually steady to the user. An intuitive example is an electronic compass: the pointer always stays at a fixed position even when the device is rotated. Similar approaches are used in VR/AR systems to keep virtual objects fixed in the user's field of view.

The motion implemented by the system is two motions that can be implemented reliably on the PIC platform: planar rotation and z-axis displacement. An orientation sensor gives information about angular velocity to estimate the heading. A pressure sensor gives an estimation of height. A manually triggered reset input sets the angle and height reference. The subsequent angle and radius deviations are processed by a discrete feedback correction loop that adjusts the angle and radius of a simple GLCD plot.

The design implemented in this project is kept simple but still demonstrates a complete sensing-to-correction loop. It also gives a clear basis for future extension work, including full 3-DOF stabilization of distance and angle, additional filtering algorithms, or more advanced correction methods.

II. High-level Designs

A. *Hardware Block*

TABLE I
Hardware Summary Table

Component	**Model Number**	**Notes (Interface / Capability / Role in Project)**
Pressure Sensor	Pressure 5 Click (BMP388)	SPI/ I^2C interface. used to measure vertical z direction variation.
Orientation Sensor	6DOF IMU 2 Click (ICM-20600)	SPI/ I^2C interface. Contains 3-axis accelerometer and gyroscope, used to detect in plane rotation.
Microchip/ microcontroller	PIC18F87K22	Handles sensor reading, control loop, and display parameter updates.
Graphic LCD Display (**GLCD**)	WDG0151-TMI-V#N00 (on PIC board)	8-bit parallel interface. display requiring manual pixel control (0/1).

The hardware used in this project presents in Table I. And hardware block diagram shown in Fig.1.Rotation and vertical motion of the IMU and pressure sensor are sensed and both signals are sent to the PIC through SPI. The control logic compares this to the user-set reference and outputs new drawing parameters to the GLCD. In this case, we can get stable and controlled visual height and orientation of the image on the screen.

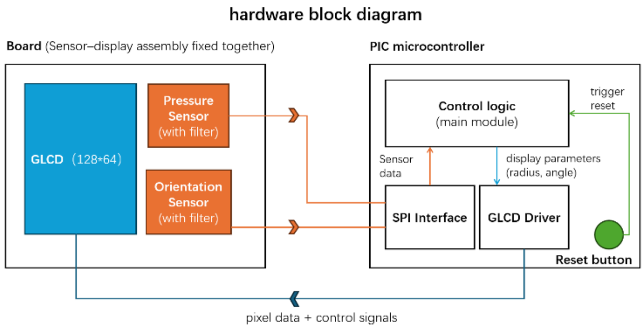

Fig. 1 hardware block diagram

The chosen sensors are sufficient for the measurable goals of the project. The pressure sensor provides a relative accuracy of ±0.08 hPa and a lowest-noise mode of 0.03 Pa (0.25 cm)[1]. With oversampling enabled, the effective height resolution is within a few centimetres, which is adequate for detecting the 15-20 cm z-axis range used in this project.

The orientation sensor supplies stable angular-rate data with very low noise densities (on the order of a few mdps/Hz) [2], allowing stable heading estimation at the control-loop rate. Any accumulated drift is removed using the manual reset input. SPI is used for communication, avoiding the addressing overhead and bus-arbitration requirements of I^2C. Also, use built-in digital low-pass (orientation) and IIR filtering (pressure), no extra capacitors is required.

The control loop runs at 20–25 Hz. The LCD panel has a slow optical response of approximately 150–300 ms[3], meaning that rapid frame updates are naturally smoothed, producing a slightly delayed but visually stable display.

B. *Software Structure (Flowchart)*

The GLCD does not include any built-in drawing primitives, so all pixels must be generated in software. For clarity and simplicity, the system displays a centre-fixed circle and a single radial line. The circle's radius represents z-axis motion, and the line's direction represents planar rotation. When the board rotates, the line rotates. When the board moves up or down, the circle grows or shrinks.

The main software design is shown in Fig.2. The software is organised into three modules: sensor interface, control logic, and GLCD drawing.

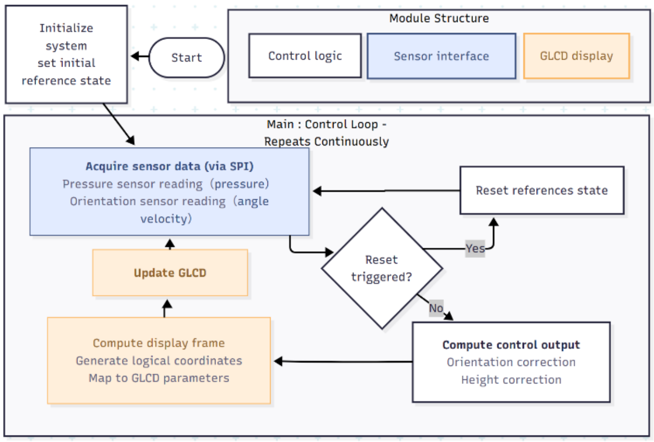

Fig. 2 main software flowchart

The software follows a simple loop structure. After initialising the sensors, SPI and the GLCD, the PIC takes one reading to set the reference angle and height. The program then enters a repeating loop: it reads the current sensor values, checks the reset input, updates the reference if the button is pressed, and computes the angle and height deviations. These deviations are turned into small corrections for the displayed line direction and circle radius. The drawing module converts these corrected values into pixel data and updates the GLCD.

The control logic uses simple discrete corrections. Angle is corrected by adding the gyros rate (angular velocity) values. The ref angle is the angle for 'zero' and each loop we increment by a small correction that pulls the plotted line back towards this ref. For height, the pressure reading is compared with the reference pressure, and this deviation is mapped to a change in radius. Again, only small incremental updates are applied each cycle. These incremental corrections work well with the GLCD's slow optical response, producing smooth visual motion without jitter.

The drawing module converts the corrected angle and radius into logic-space coordinates. A simple circle routine and a Bresenham-style line routine generate the pixel data. The results are then mapped to the GLCD's page and column format. Each frame overwrites the previous one, giving a stable plot.

There are several extensions we could add to the basic loop. The control logic could be extended in several directions. More advanced correction methods, such as derivative damping or anti-windup could be added. The IMU also supports roll and pitch, enabling a future upgrade to full 3-axis stabilisation with a complete reference frame, like drone initialisation procedures. Additional software filtering could be used if it essential.

III. Projected Performance and Assessment Criteria

A. Angular and Distance Visual Stability

The system aims to keep the display stable during normal hand-held motion. For rotation, the target is to keep the drift of the radial line within 10-20°. For height, the radius error of circle is set to 1–3 cm. It is suitable for the ideal case of 40 – 50 cm viewing height and for z-axis change on relatively small screen of 128×64. Both above values are in an acceptable range of view during short time and with moderate movement of hand.

The above two targets will be checked with controlled motions. For the board rotation, we will rotate it to known angle (e.g. 30º, 45º, 90º) and move it vertically by 15 – 20 cm downward/upward approximately uniformly. We will check both the recording and the output value. The recording allows us to have a perception of the stability. The output value is compared with the ideal case of uniform motion, which is used to check whether the deviation is within the above chosen range.

B. Correction Loop Responsiveness

The correction loop is expected to react within 2-4 control cycles, corresponding to roughly 100-150ms at a 20 Hz loop rate. This range is appropriate for an incremental correction method and matches the response limitations of the GLCD(3-7Hz).

While Section A discussed the final stability of the display, this section will test the system how quickly can it react when it is disturbed. As we input a sudden change of angle and a small change of height into the loop, it will be fed into the loop, and we can check how quickly can the values start to move back to the reference. This provides a consistent reference case that can be compared with the real tests and helps identify the range that the system can no longer settle within a few cycles.

C. System Timing and Operating Range

The system is designed to run with a consistent loop time of 40–50 ms, giving an effective update rate of 20-25 Hz. This rate is chosen because it matches the combined duration of the SPI sensor reads, the correction step and the GLCD redraw. The rate will be checked with a debug pin and the period is measured by an oscilloscope to make sure the timing of update is stable in a range.

The operating range of the system will also be characterised to reveal its limitations. As we apply controlled change of angle and change of height in section B, we can obtain an estimate of rotation speed and z-axis behaviour of change that the system can track.

In addition, extended tests without resetting will indicate how much cumulative drift builds up after several rotations or repeated vertical movements, showing its practical limits. The measurement will show the behaviour of the system beyond the main stability and responsiveness.

IV. Conclusion

This project demonstrates a practical design of a low-dimensional visual stabilization system. By combining SPI-based sensor acquisition, a discrete controller, and real-time LCD rendering on the PIC18 platform, the system achieves stable compensation for x-y plane rotation and limited z-axis motion. The defined performance targets, a drift tolerance of 10-20° and a radius error of 1-3 cm at a 20-25Hz loop rate are achievable with the chosen sensors and software structure. the design is practical within the constraints of an 8-bit microcontroller.

Looking forward, the approach can be extended to full 3-axis orientation control, improved display mapping, and more advanced filtering or control methods. These extensions would address the current drift and operating range limitations and would enable the framework to scale up to more capable stabilization tasks.

References

[1] Bosch Sensortec, *BMP388 Digital, Barometric Pressure Sensor – Datasheet*, pp. 1–2 (absolute/relative accuracy, noise), 2017.

[2] TDK InvenSense / MikroElektronika, ICM-20600 6-axis IMU – Gyroscope and Accelerometer Specifications, accessed 2025.

[3] Winstar Display Co., *Graphic LCD Module WDG0151-TMI-V#N00 – Specification*, pp. 6–7 (optical response time), 2009.

APPENDIX

Feedback statements:

The 2D plot showing your data and the large backgrounds from the resonances was powerful and illustrative. Although, it doesn't appear that the red box corresponds to your fit region.

Something that can be improved for your next experiment is to be more descriptive about what you did and why. In a presentation this short some steps need to be glossed over, but some are needed to give confidence in your results. For example, where did the shapes for your signal and background components come from, and how confident you are that they are good enough models.

In addition to these comments, please also refer to the statements on the rubric for both the awarded grade and the one directly above.

If you would like to briefly discuss the feedback with the 1st marker, please contact them (you can ask me if you are not sure who they were) and arrange a meeting - this will probably be during a lab session when they are demonstrating. This is not an opportunity to challenge the grades awarded, however. If, after this meeting, you feel the awarded grades are incorrect, please email me detailing which grade(s) and reasons why.